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Effect of the RN⁻ Gene on Ultrastructure and Protein Fractions in Pig Muscle

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ABSTRACT

The aim of the present experiment was to study the effect of the RN⁻ gene on ultrastructure of white fibres and the proportion of the different protein fractions in Longissimus lumborum *muscle of RN⁻ pigs. The ultrastructure of muscle was observed in 4 RN⁻ carrier pigs and 4 rn⁺rn⁺ homozygous pigs by mean of electron microscopy. Protein fractionation and quantification were performed on muscle from a second group of 4 RN⁻ and 4 rn⁺ pigs. No change in the proportions of the different protein compartments was observed. The abnormality of the glycolytic fibres of RN⁻ pigs seems to be rather due to an increase of the glucid concentration in the sarcoplasm, perhaps secondarily to a modified osmolarity in the intracellular medium.*

INTRODUCTION

The dominant RN⁻ gene affects the meat quality of pigs by increasing the glycogen content of muscle and by inducing the production of 'acid meat' (Monin, 1989; Fernandez *et al.*, 1992). The glycogen level is increased by about 70% in white muscle of RN⁻ carrier pigs (RN⁻ pigs) as compared with homozygous rn⁺rn⁺ pigs (rn⁺ pigs). The glycogen excess in muscle from RN⁻ pigs is mainly observed in glycolytic fibres (Estrade *et al.*, 1992). Preliminary studies indicated that the distribution of glycogen particles is comparable in tissues from both RN⁻ and rn⁺ pigs, but both amount and density of the glycogen particles were seen to be higher

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in glycolytic fibres of RN^- pigs as compared with rn^+ pigs. Moreover, the ultrastructure of glycolytic fibres differs between RN^- and rn^+ pigs, fibres of RN^- animals showing abnormally large sarcoplasmic spaces between myofibrils (Estrade *et al.*, 1993). It remains unclear whether this morphological alteration is primarily related to a difference in the proportion of the protein compartments (myofibrillar, sarcoplasmic, connective), or if it is the consequence of an increase in the glucid concentration of the sarcoplasmic compartment.

The aim of this study was to test these hypotheses by estimating the respective proportions of stromal, sarcoplasmic and myofibrillar proteins, and by assessing the concentration of glycogen in the sarcoplasmic compartment in comparison with the sarcoplasmic protein concentration.

MATERIAL AND METHODS

Animals

The glycolytic potential was determined on muscle biopsy samples from 50 pigs weighing about 70 kg of an experimental herd including RN^- pigs. Eight RN^- pigs with a glycolytic potential between 291 and 357 μmol/g fresh muscle and 8 rn^+ pigs with a glycolytic potential between 161 and 201 μmol/g fresh muscle were retained for the study. The animals were slaughtered in pairs at around 100 kg liveweight by electronarcosis and exsanguination in an experimental slaughterhouse, each pair including 1 RN^- and 1 rn^+ pig. These animals were divided in two groups (8 animals), one devoted to electron microscopic studies, the other to the biochemical studies.

Electron microscopy

Muscle samples were taken by shot biopsy (Talmant *et al.*, 1989) for microscopic study from 8 pigs, i.e. 4 RN^- and 4 rn^+ pigs.

Small blocks (1–2 mm^3) were cut from the muscle biopsies, fixed for 2 h at 4°C in 2·5 % glutaraldehyde in 0·1 M cacodylate buffer pH 7·2 and post-fixed in 1% osmium tetroxide in the same buffer for 1 h at 4°C. The specimens were dehydrated through an ethanol gradient and embedded in epoxy resin.

Ultrastructural morphology and glycogen particles were observed on ultrathin sections stained by the periodic acid–thiocarbohydrazide (TCH)–silver proteinate method (PATAg) according to Thiéry (1967). The observations were made with a Philips EM 400 electron microscope at an accelerating voltage of 80 kV.

Biochemical study

Animals and sampling

Eight animals (4 RN^- and 4 rn^+) were involved in the biochemical study. Within 5 min after slaughter, 100 g of *Longissimus lumborum* were removed from the animals and put on ice. A 5 g sample of the muscle was immediately taken and dipped in liquid nitrogen. It was then freeze-dried and subsequently used for glycolytic potential determination. The muscle (40 g) was dissected free of fat and connective tissue and minced with a domestic mincer. Approximately 5 g of ground muscle were frozen at −20°C and freeze-dried. They were powdered in a coffee-mill before use for total protein determination.

Longissimus lumborum muscle (10 g) was immediately homogenized in 9 vol of phosphate buffer (0·03 M, pH 7·4) for protein fractionation.

Analytical techniques

Glycolytic potential determination. Glycogen was estimated as glycolytic potential, i.e. Glycolytic potential = 2 ([glycogen] + [glucose] + [glucose-6-P]) + [lactate] determined from about 250 mg of freeze-dried tissue as described by Monin & Sellier (1985). The amounts of glycogen + glucose + glucose-6-phosphate were determined using the method of Dalrymple & Hamm (1973), lactic acid was determined according to Bergmeyer (1974).

Protein fractionation. The muscle proteins were fractionated according to the method of Hellander (1947). The fractions (sarcoplasmic protein, myofibrillar protein, stromal protein and non-protein) were freeze-dried before determination of nitrogen.

Protein determination. Nitrogen was determined after Kjehldal mineralization according to the technique of Ferrari (1960). Protein content was calculated as total nitrogen × 6·25.

RESULTS

Electron microscopy

After PATAg staining, glycogen appeared as distinct particles in the sections (Fig. 1, (1) and (2)). Glycogen was localised mainly in intermyofibrillar and subsarcolemmal spaces. The ultrastructure of glycolytic fibres differed between RN^- and rn^+ pigs. The glycolytic fibres of rn^+ pigs showed narrow and regular intermyofibrillar spaces as usually observed in normal muscle (Fig. 1(1)). In muscle from RN^- pigs, the glycolytic

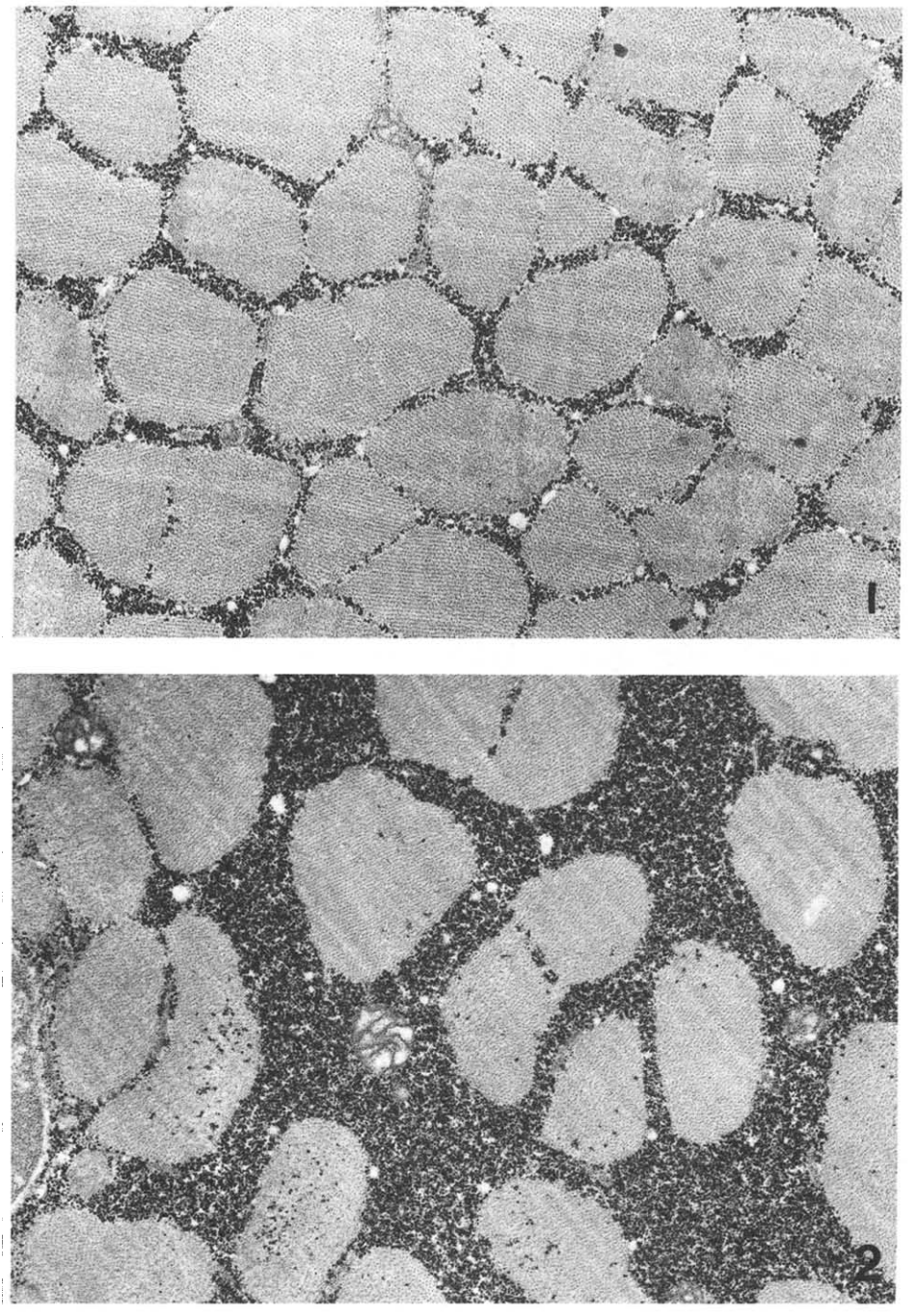


Fig. 1. Transverse sections of glycolytic fibres from *Longissimus lumborum* biopsies: (1) muscle of rn^+ pig; (2) muscle of RN^- pig. Note the difference in the size of the sarcoplasmic compartment between rn^+ and RN^- muscle (× 14 500).

TABLE 1
Effect of the RN^- Gene on Glycolytic Potential (GP) and Total Protein of *Longissimus lumborum* Muscle

	RN^-		*rn^+*
GP (μmol/g)	335 ± 30	**	187 ± 15
Total protein (mg prot/g)	206 ± 5	**	225 ± 4
GP/total protein	1·6 ± 0·2	**	0·8 ± 0·1
GP/sarc. protein	5·6 ± 0·3	**	3·0 ± 0·3

Results are means ± sd. Significant differences between RN^- and rn^+ pigs: ** $P < 0.01$.

fibres showed an abnormally large sarcoplasmic compartment. The latter was broad and irregular in shape, and the section of the myofibrils was very variable (Fig.1(2)). In both cases, glycogen particles were abundant in all the sarcoplasm, but seemed to be at a higher density in RN^- muscles.

Analytical results

Glycolytic potential was higher by about 70% in muscle from RN^- pigs as compared with m^+ pigs (Table 1). By contrast, the total protein content was lowered by about 10% in muscle from RN^- as compared with rn^+ pigs (206 and 225 mg protein/g of fresh muscle, respectively). The ratio of glycolytic potential to total protein was higher in muscle from RN^- pigs than in muscle from m^+ pigs.

Table 2 shows the proportion of the different protein fractions. The difference observed for the total protein content between RN^- and rn^+ pigs was seen in all the protein compartments. Hence, the distribution of the protein fractions was similar in RN^- and rn^+ pigs.

TABLE 2
Proportion of the Different Protein Fractions in Muscle from RN^- and rn+ pigs

	Total nitrogen (%)	
	RN^-	*rn^+*
Myofibrillar protein	47 ± 3	46 ± 2
Sarcoplasmic protein	32 ± 1	31 ± 3
Stromal protein	4 ± 2	6 ± 2
Non-protein nitrogen	17 ± 2	16 ± 2

Results are means ± sd.

DISCUSSION

The glycolytic potential was higher (+70%) and the total protein content was lower (−10%) in muscle from RN^- pigs than in muscle from rn^+ pigs. This result is in good agreement with those of Monin & Sellier (1985), Monin *et al.* (1986) and Monin *et al.* (1992), who reported similar observations in Hampshire and Hampshire crossbred pigs, which could be considered as RN^- pigs, as discussed by Monin (1989).

The present results clearly show that, despite the difference in total protein content, the proportion between the protein fractions is similar in muscles from RN^- and rn^+ pigs. Therefore the difference in the glycolytic fibre ultrastructure between RN^- and rn^+ pigs appears not to be linked to a change in the proportions of the different protein compartments. The abnormal ultrastructure of the glycolytic fibres from RN^- pigs is likely to be due to an increase in the glucid concentration in the sarcoplasm, leading to the observed increase in the relative volume of this compartment. These observations strongly suggest that the RN^- gene is associated with an abnormality in the glycogen metabolic pathway. However, another hypothesis can be put forward. Baquet *et al.* (1990) and Li *et al.* (1992) studied the effect of the medium osmolarity on the glycogen content in liver and muscle. They showed that osmolarity may be an important determinant of tissue glycogen content. Several amino acids are known to stimulate glycogen synthesis via activation of glycogen synthase. This stimulation is due to an increase in hepatocyte volume resulting from the uptake of these amino acids. These authors demonstrated that cell swelling *per se* stimulates glycogen synthesis. The increase of the sarcoplasmic compartment volume in the glycolytic fibres of RN^- pigs could be associated with a modified osmolarity of the cell, inducing an increase in intracellular water. It is therefore possible that the glycogen accumulation seen in the glycolytic fibres of RN^- pigs occurs secondary to a swelling of the sarcoplasmic compartment. This hypothesis is in good agreement with the result of Monin *et al.* (1992), who has reported that the ratio of water to protein is higher in RN^- pigs as compared to normal pigs.

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